

Election Management System - Functional Requirements Document

System Overview

The Election Management System is a comprehensive web application built with React, Node.js, and MongoDB, designed to facilitate the management of multiple elections across different franchises. The system supports multiple user permission levels, election creation and management, nominee handling, voter management, and detailed election analytics.

User Roles and Permissions

1. Franchise Admin

- Create unlimited elections
- Manage franchise details (logo, settings)
- Create/manage election admins
- Access analytics across all elections
- Import/export election data
- Configure election parameters
- Create and manage reusable election groups

2. Election Admin

- Manage specific assigned elections
- Add/remove nominees
- Generate and manage voters
- View election results and analytics
- Monitor voting status

3. Voter

- Self-register (if enabled)
- Login with credentials
- Cast votes in assigned elections
- View confirmation of vote submission

Functional Requirements

1. Franchise Management

- **FR1.1:** System shall allow creation of multiple franchises
- **FR1.2:** Each franchise shall have unique identification and settings

- **FR1.3:** Franchise admins can customize branding with logo uploads
- **FR1.4:** Franchise settings shall include default parameters for elections

2. Election Configuration

- **FR2.1:** Create elections with configurable parameters:
 - Organization name
 - Election title
 - Election date
 - Number of positions to be elected
 - Nominee display order
 - Maximum voters allowed
 - Maximum nominees allowed
 - Gender requirements (male/female minimums)
 - Self-registration options
 - Voting status (open/closed)
- **FR2.2:** Election settings shall be editable before election start
- **FR2.3:** System shall support election templates/groups for reuse

3. Nominee Management

- **FR3.1:** Add nominees individually with name and other details
- **FR3.2:** Bulk upload nominees via CSV/Excel
- **FR3.3:** Copy nominees from one election to another
- **FR3.4:** Edit and remove nominees before election starts
- **FR3.5:** Organize nominees in specified display order

4. Voter Management

- **FR4.1:** Generate voter accounts individually with:
 - Username
 - Password
 - Registration number
- **FR4.2:** Bulk create voters with:
 - Configurable prefix
 - Starting number
 - User count
- **FR4.3:** Print user credentials list
- **FR4.4:** Export voter details to Excel
- **FR4.5:** Edit individual voter details
- **FR4.6:** Group voters for reuse in multiple elections

5. Voting Process

- **FR5.1:** Toggle voting open/closed status
- **FR5.2:** Secure voter authentication
- **FR5.3:** Present ballot with configured nominees
- **FR5.4:** Validate votes against election rules
- **FR5.5:** Confirm vote submission to voters
- **FR5.6:** Prevent duplicate voting

6. Results and Analytics

- **FR6.1:** Real-time voting status display
- **FR6.2:** Track pending voters
- **FR6.3:** Count and display votes
- **FR6.4:** Sort results in ascending/descending order
- **FR6.5:** Detailed analysis of who voted for whom (admin only)
- **FR6.6:** Export results to Excel
- **FR6.7:** Print election results
- **FR6.8:** Show status without revealing vote details (configurable)

7. System Administration

- **FR7.1:** User authentication and authorization
- **FR7.2:** Audit logging of all system activities
- **FR7.3:** Backup and restore functionality
- **FR7.4:** System configuration settings
- **FR7.5:** Manage user permissions and access

Non-Functional Requirements

1. Security

- **NFR1.1:** Secure authentication with password hashing
- **NFR1.2:** Role-based access control
- **NFR1.3:** Data encryption for sensitive information
- **NFR1.4:** Protection against common web vulnerabilities

2. Performance

- **NFR2.1:** Support concurrent users during peak voting periods
- **NFR2.2:** Response time < 2 seconds for standard operations
- **NFR2.3:** Support for large elections (1000+ voters)

3. Usability

- **NFR3.1:** Intuitive user interface for all user roles
- **NFR3.2:** Mobile-responsive design
- **NFR3.3:** Clear error messages and user guidance

4. Availability

- **NFR4.1:** System uptime of 99.9% during active elections
- **NFR4.2:** Graceful handling of connection issues

5. Scalability

- **NFR5.1:** Support for multiple simultaneous elections
- **NFR5.2:** Horizontal scaling capabilities